

Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

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Overview

Introduction of retinal prosthetics

Encoding strategies

- Reverse encoding
- End-to-end encoding
- Invertible encoding

Eye gaze

- Visual fixation-based simulation
- Visual fixation consistency

Supplementary

- Questionnaire
- Braille simulation

B. Epiretinal

C. Subretinal

Nerve fibers
to optic
nerve

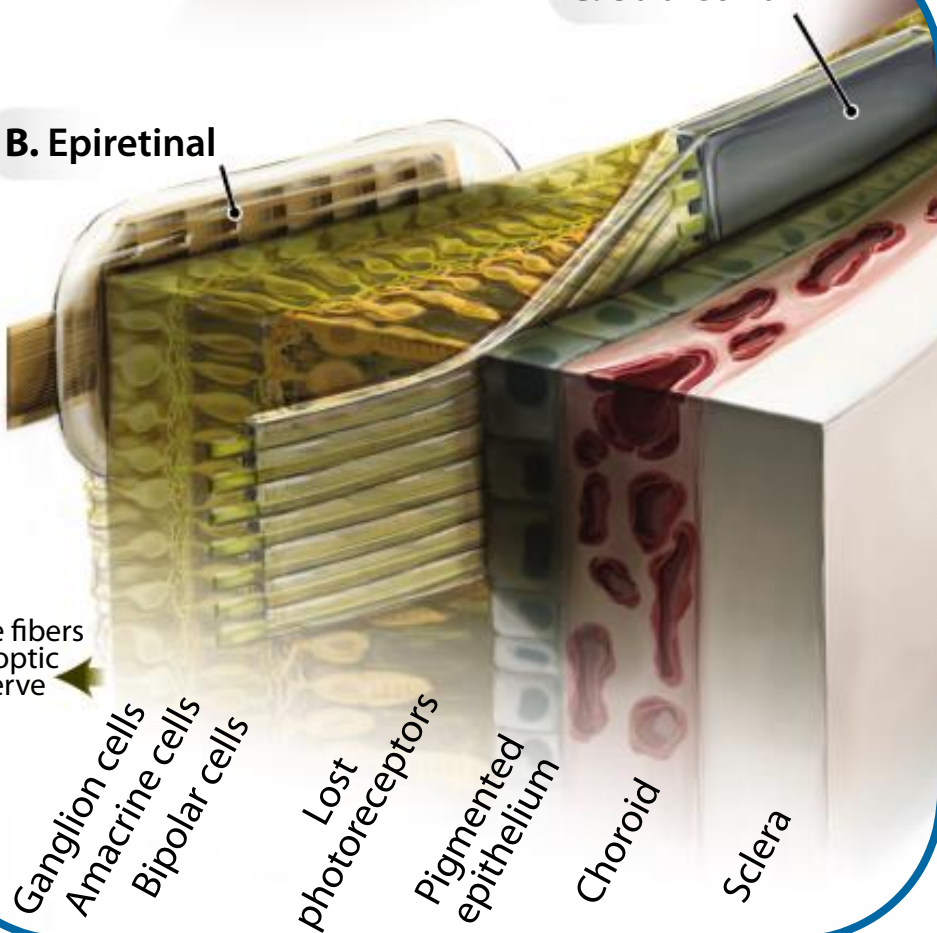
Ganglion cells
Amacrine cells
Bipolar cells

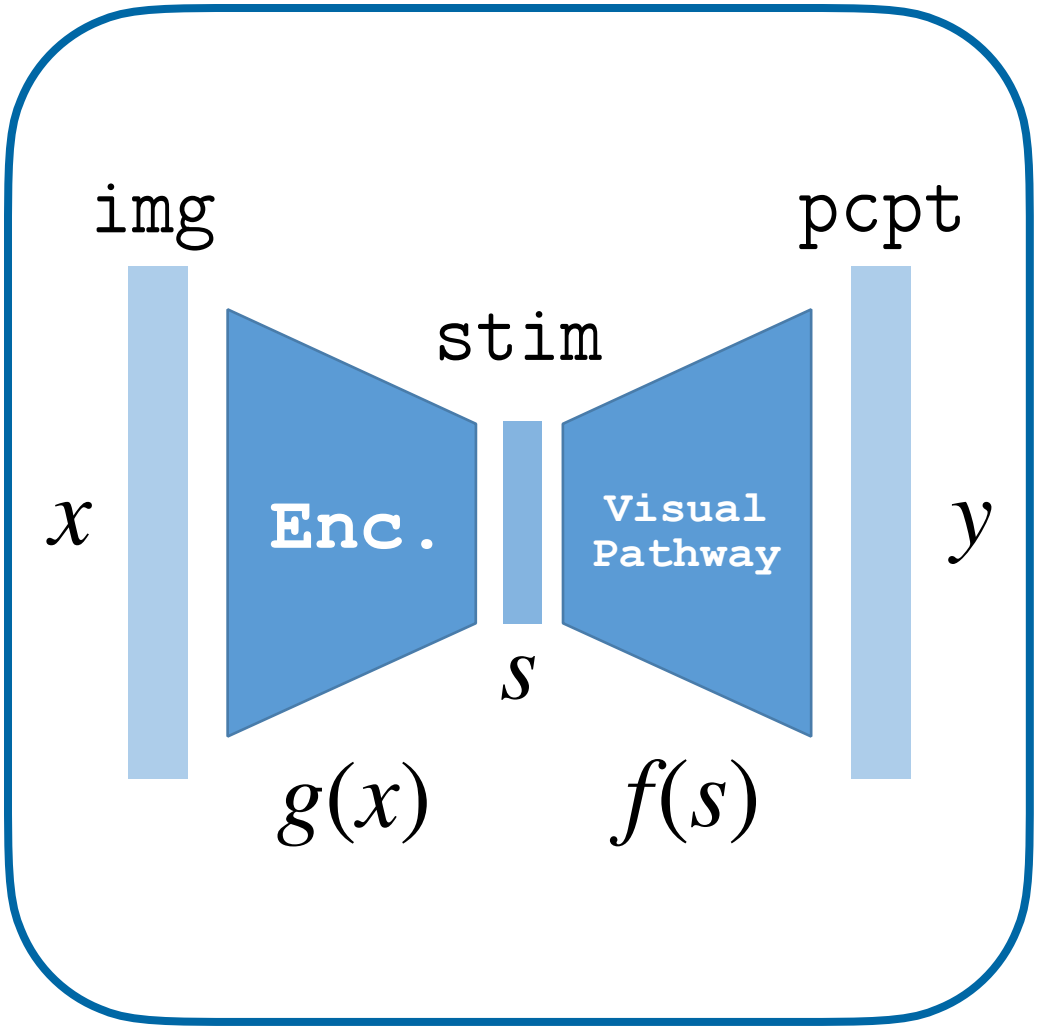
Lost
photoreceptors

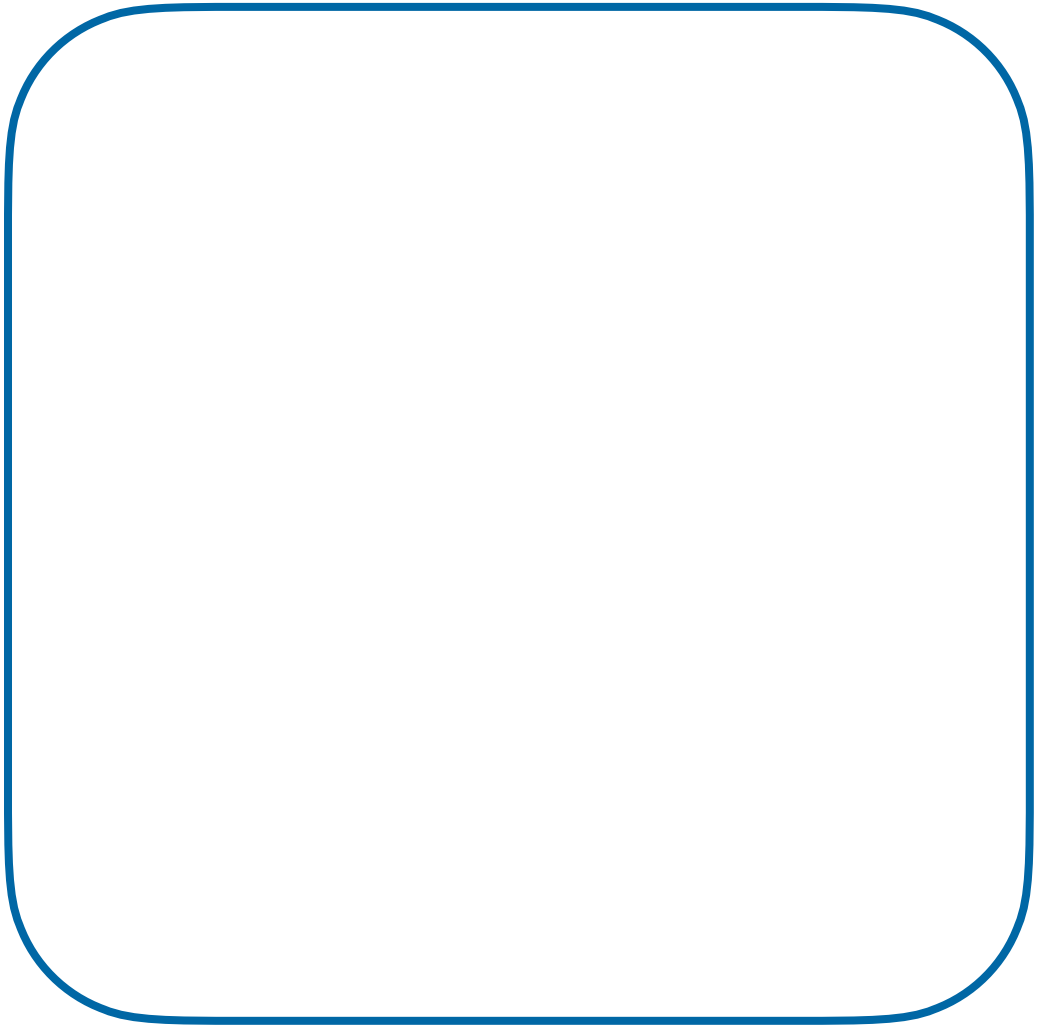
Pigmented
epithelium

Choroid

Sclera

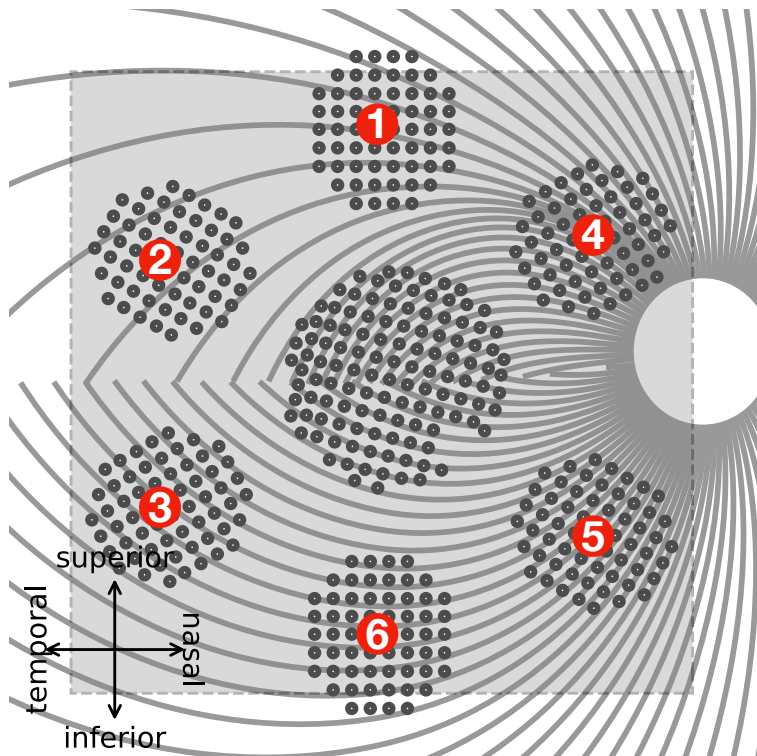






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0.00s



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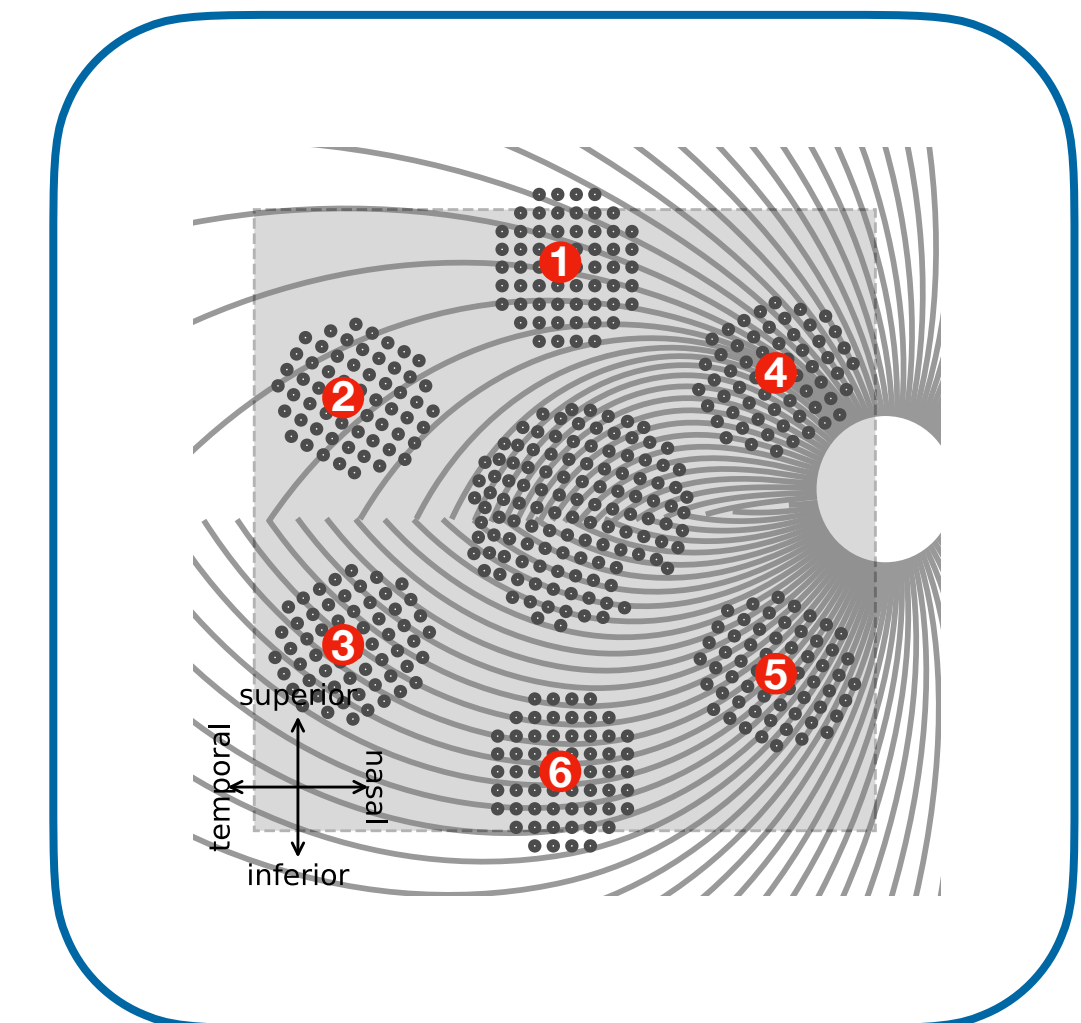
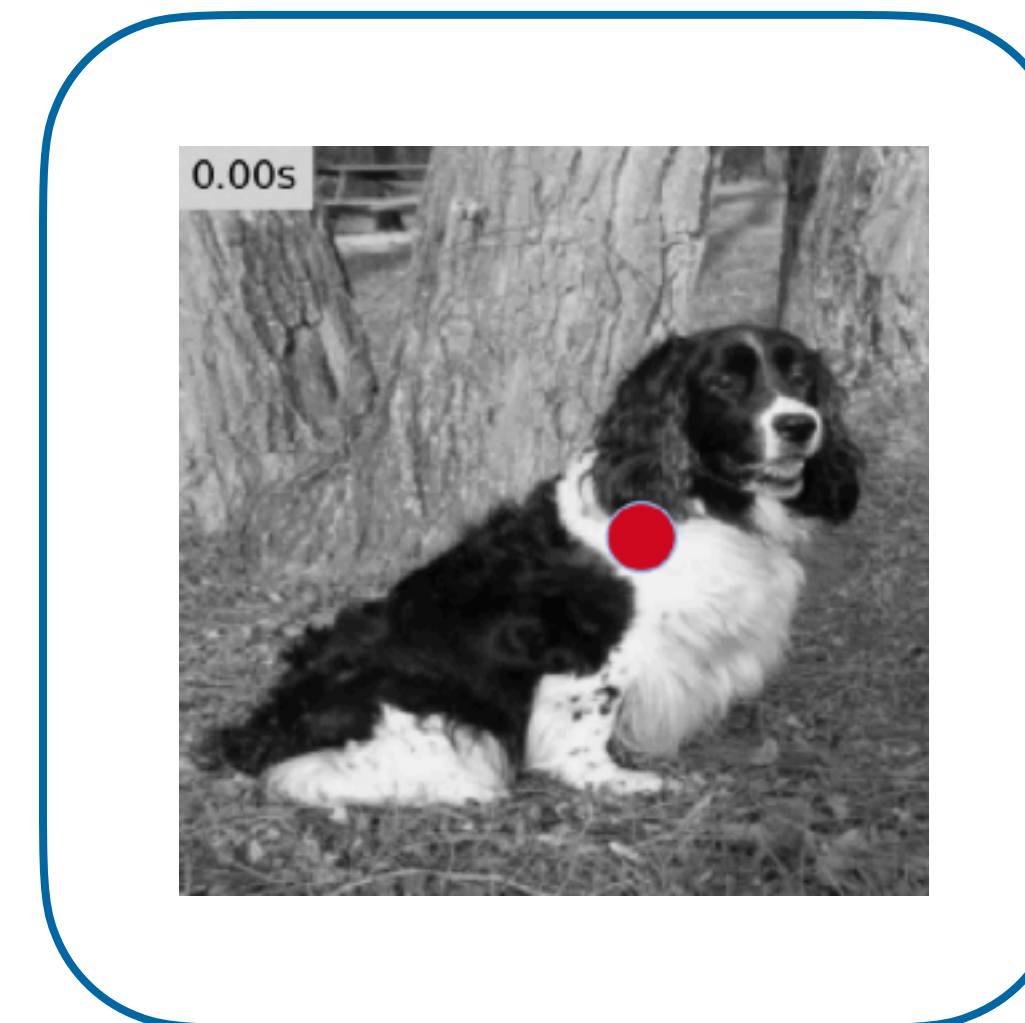
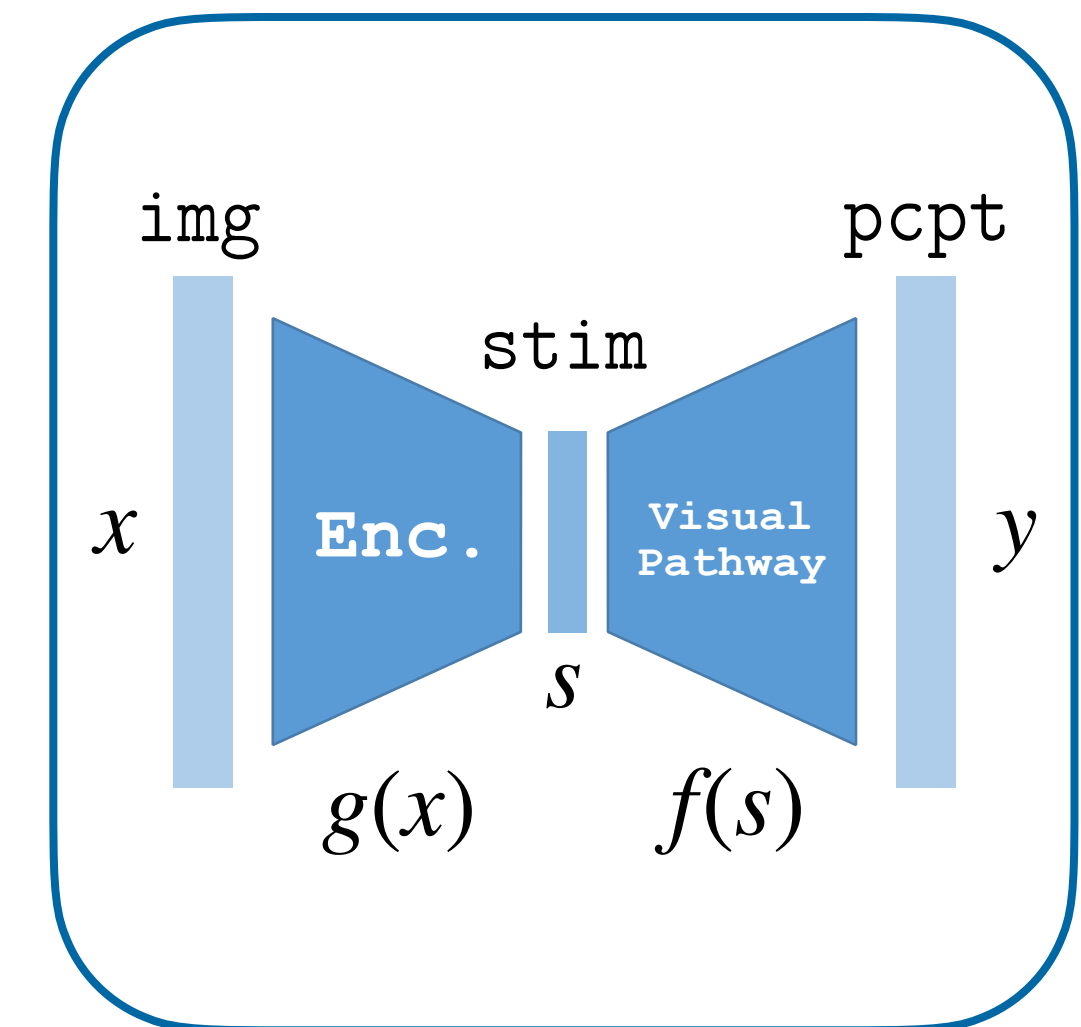
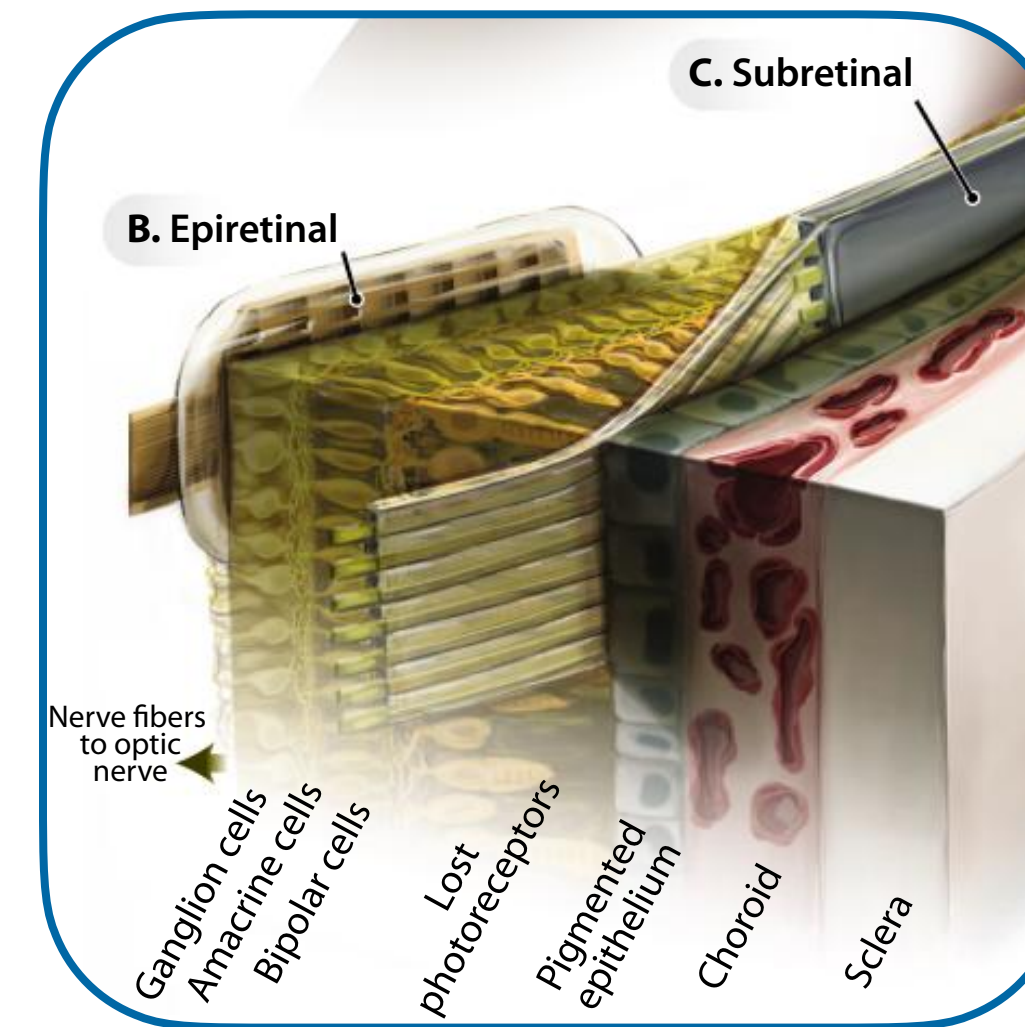
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- Invertible encoding

Eye gaze

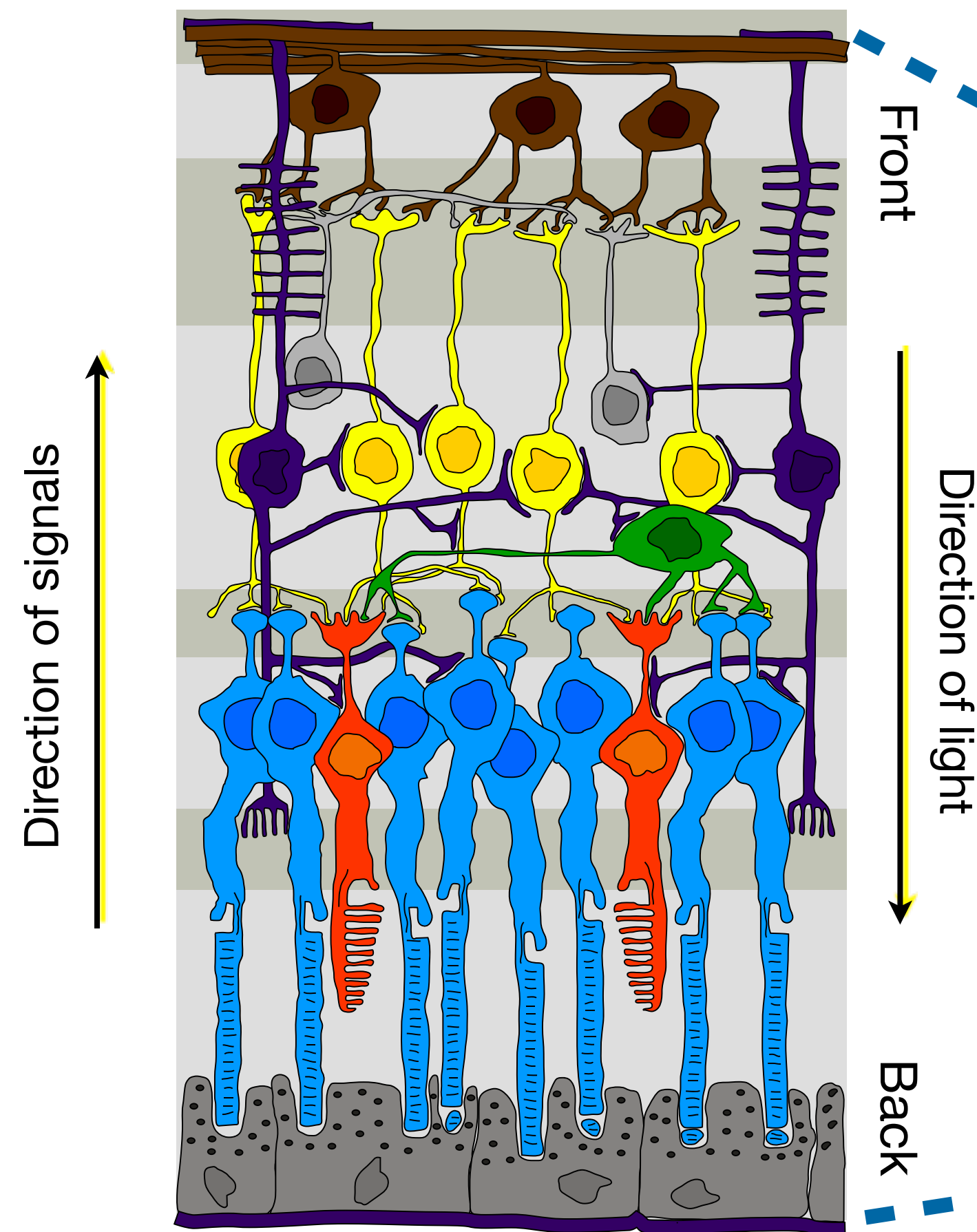
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Supplementary

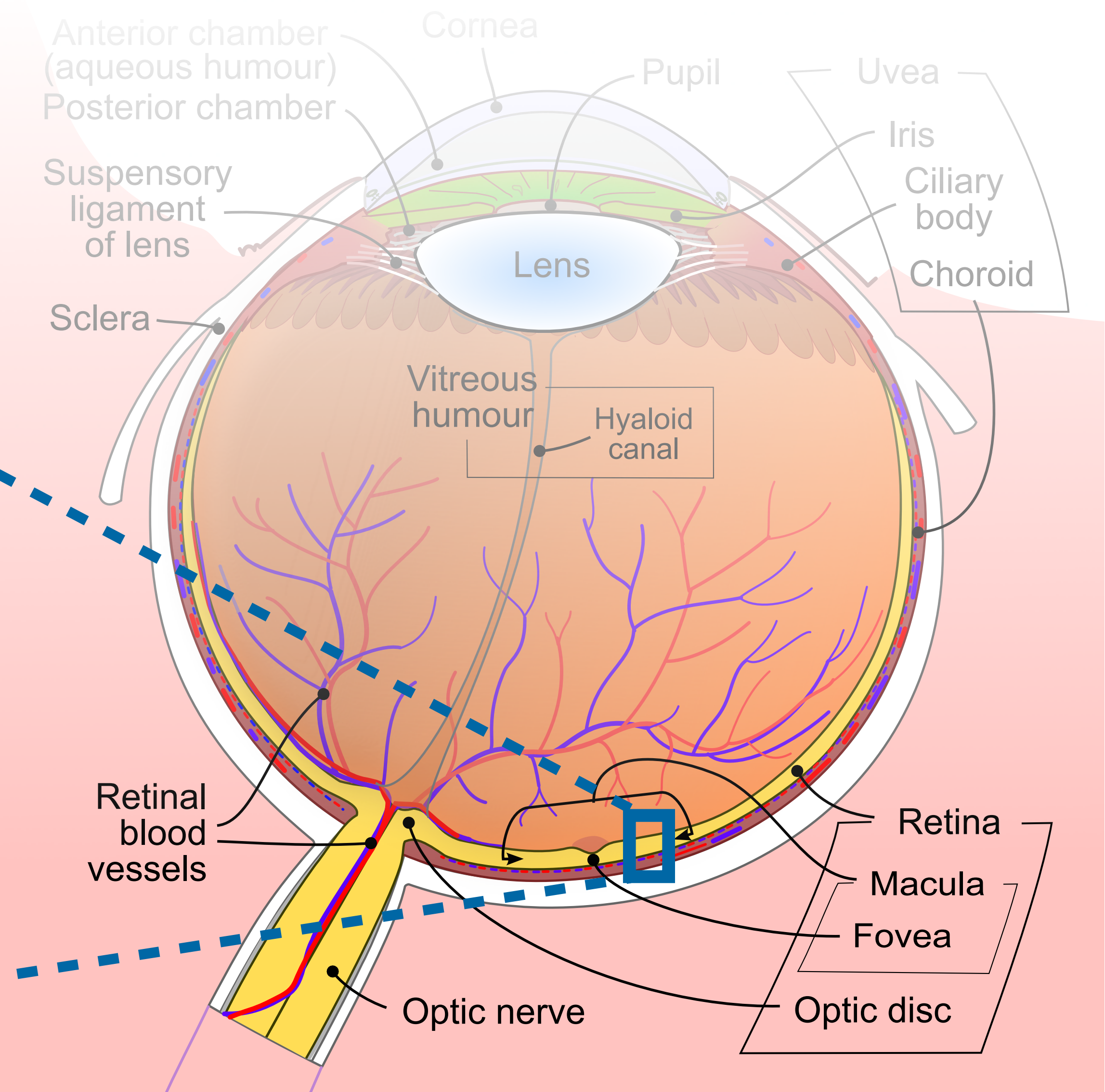
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Human Retina



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Rhcastilhos & Jmarchn, Schematic diagram of the human eye
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